A General Theory of Selection: Two-Stage Sieves, the Information Profit Threshold, and Cyclotomic Algebraic Substrates

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Abstract

Selection from astronomically large candidate spaces recurs across domains as disparate as subatomic physics, molecular biology, and corporate finance. We propose the **General Selection Principle** (GSP): the observed configuration is selected by the conjunction of two independently motivated filters—admissibility (structural/arithmetic consistency) and viability (a dynamical threshold on generative-to-dissipative rates).

The viability filter requires the rate ratio to exceed the *Information Profit Threshold* $\mathrm{IPT} = 1 + \frac{\ln\varphi}{2\ln(2\pi)} \approx 1.1309$, derived from first principles in [1]. Together, the filters produce asymptotically sparse but non-empty survivor sets.

We document two independent quantitative signatures of the GSP. The IPT threshold recurs across five empirical domains — ecology and economics (prior derivation [1]), nuclear magic numbers, prebiotic chemistry, corporate survival, and microbial metabolism (FBA, $r = 0.971$, $p < 0.001$) — together with one model-internal symbolic-communication result (natural language alphabet size). The algebraic substrate $\mathbb{Q}(\zeta_{120})$ recurs in the Standard Model parameter spectrum [2], the Zamolodchikov E8 integrable QFT mass ratios [3], ADE Toda field theories, and $\mathrm{SU}(2)_k$ WZW quantum dimensions. The E7 Toda theory ($h = 18$, $18 \nmid 120$) provides a clean falsifier: all six mass ratios lie provably outside $\mathbb{Q}(\zeta_{120})$, with the exclusion Lean-certified [3].

Evidence grades range from [T] (theorem-level, machine-certified) through [B] (empirical, replication pending) to [B$^-$] (empirical, single-laboratory, not yet independently replicated); [Br] denotes bridge claims conditional on companion-paper premises; a domain-by-domain evidence map appears in Table 1. The co-occurrence of these two independent signatures supports the hypothesis that a single mathematical mechanism underlies selection across discrete combinatorial systems; see also the companion Self-Referential Renormalization Group (SRRG) paper [4] for a proposed mechanism.

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Claim Status Taxonomy (used throughout this paper)	
[T]	*Theorem*: machine-checked in Lean 4 (`ugp-lean`), zero `sorry`
[C]	*Computational*: verified by deterministic script at stated precision
[B+]/**[B]**/**[B$^-$]**	*Empirical* evidence (decreasing strength: independent replication / single dataset / single-laboratory, not yet replicated)
[Br]	*Bridge claim*: conditional on companion-paper premises
[I]	*Interpretive*: plausible reasoning; not yet formally verified
[Conj]	*Conjecture*: clearly stated open hypothesis

1 Introduction

Among the most striking features of nature is the extreme sparsity of realised structures relative to the space of conceivable alternatives. The Standard Model of particle physics selects a unique gauge group and generation content from a vast landscape of possible theories. Biological evolution converges on a single genetic code from a combinatorial space of $\sim 10^{84}$ possible codon assignments. Nuclear shell structure exhibits magic numbers at exactly seven values despite the continuous spectrum of available energy levels. In each case, the question arises: what principle, if any, underlies the selection?

A partial answer has emerged from two independent lines of work within our research programme. The first, developed in [1], identifies the **Information Profit Threshold** IPT ≈ 1.1309 as the critical ratio of generative to dissipative rates at which a self-referential system transitions from non-viable to viable. This threshold is confirmed in ecological, economic, and population-dynamics data. The second, developed in [2], establishes $\mathbb{Q}(\zeta_{120})$ as the algebraic substrate for the Standard Model parameter spectrum: all Universal Generative Principle (UGP)-derived constants lie in this cyclotomic field.

The present paper asks whether these two signatures are domain-specific artefacts or instances of a general law. We test each against three additional physical systems — nuclear magic numbers, the genetic code, and integrable quantum field theories — and find consistent evidence that both signatures extend beyond their original context.

1.1 The two-stage sieve mechanism

The pattern common to the systems we study was identified and formalised in companion papers P01 [2] and P24 [5] of this programme, building on the UGP framework [6, 7]. In each domain, the observed structure emerges through two successive filters:

1. **Admissibility** (Stage 1): a structural consistency condition — arithmetic, geometric, or biochemical — that eliminates the vast majority of candidates. Admissibility is domain-specific in form but appears to have a common effect across these domains: it reduces the candidate space by many orders of magnitude.
2. **Viability** (Stage 2): a dynamical threshold condition. Among admissible configurations, only those whose generative-to-dissipative rate ratio exceeds IPT survive indefinitely. Non-viable configurations decay or are outcompeted.

The intersection of these two filters yields an asymptotically sparse but non-empty survivor set — the *selected* structures.

Definition 1.1 (Two-stage sieve)**.** *Let $\mathcal{C}$ be a discrete space of candidate configurations. Let $A \subseteq \mathcal{C}$ be the admissible set (Stage 1) and $V \subseteq \mathcal{C}$ the viable set (Stage 2). The* survivor set *is $S = A \cap V$.*

Theorem 1.2 (Asymptotic Sparsity [**Br**]; full proof in [2, 6])**.** *Under generic conditions on A and V, $|S|/|A| \to 0$ as $|\mathcal{C}| \to \infty$.*

*Proof sketch [**Br**].* The full proof is given in [2] (Theorem 2.1) and [6]. The concrete UGP instance — the joint Stage-1/Stage-2 sieve has exactly one solution ($n = 10, b_1 = 73$) across all ridge levels $n \in \mathbb{N}$ — is Lean-certified [**T**]: `asymptotic_sparsity_universal` (module `Phase4.AsymptoticSparsity`, zero `sorry`, [3]).

The *general* IPT form of this theorem is now also Lean-formalized in `GXT.AsymptoticSparsity` [3] (GXT = Geometric Threshold, the conjecture that the viability threshold equals IPT universally across discrete combinatorial domains): the module proves that for any N-component system with i.i.d. efficiency ratios $\theta_i \in [0, b]$ and mean $\eta_0 <$ IPT, the fraction satisfying $\theta \geq$ IPT is bounded by $e^{-\delta N}$ with $\delta = 2(\mathrm{IPT} - \eta_0)^2/b^2 > 0$, hence $\to 0$ as $N \to \infty$. The algebraic decay theorems are zero sorry; the probabilistic bound awaits Mathlib's Hoeffding inequality. **Status: [Br] (Lean)** — conditional on the Hoeffding bound hypothesis.

The key intuition is as follows. Fix the admissible fraction $\alpha = |A|/|\mathcal{C}|$; as $|\mathcal{C}| \to \infty$, structural consistency conditions generically force $\alpha \to 0$ (the admissibility filter eliminates exponentially more configurations as the search space grows). The viable set V is defined by a continuous threshold condition on the generative-to-dissipative ratio: $V = \{c \in \mathcal{C} : \theta(c) \geq \mathrm{IPT}\}$. Under the generic assumption that the distribution of $\theta(c)$ over A is absolutely continuous with finite mass at the threshold, the intersection $|A \cap V|/|A|$ tends to zero as the filter becomes increasingly fine-grained with growing $|\mathcal{C}|$. The result establishes that selection is *inevitable* (the sieve cannot be avoided as the candidate space grows) while the survivor set remains *non-empty* because the threshold IPT is defined as the unique fixed point of the Gen/Drain renormalisation group [1] — the one value at which marginal stability is achieved. □

The force of the GSP is not the general sparsity result, which is expected, but the *quantitative* recurrence of IPT ≈ 1.1309 and $\mathbb{Q}(\zeta_{120})$ across structurally unrelated domains. These two constants are derived, not adjusted; their appearance in new contexts is a falsifiable prediction.

2 The Information Profit Threshold across domains

The Information Profit Threshold is defined in [1] as the unique fixed point of the Gen/Drain renormalisation group: the critical ratio θ^* such that a self-referential system is marginally stable. Three structural premises (derived in [1]) yield $\mathrm{IPT} = 1 + \ln\varphi/(2\ln(2\pi)) \approx 1.1309$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The balance ratio $\Lambda = \ln\varphi/\ln(2\pi) \approx 0.2618$ embedded in IPT is independently identified by Norfleet [8] as a fundamental balance constant of discrete–continuous transitions, providing an independent geometric derivation of the same threshold. Below the threshold ($\theta <$ IPT), systems cannot sustain coherent self-referential processing and decay; above it ($\theta \geq$ IPT), systems are viable and self-maintaining. In a large configuration space the fraction of candidates naturally achieving $\theta \geq$ IPT is asymptotically small (see Theorem 1.2 and §2 below), so the intersection of the admissibility and viability filters yields a sparse but non-empty survivor set — the IPT is the unique fixed point at which marginal stability is achieved.

Logical relation to the companion SRRG paper. The present paper is empirical and comparative: it asks whether two independently specified signatures—the information-profit threshold

Table 1: Evidence map for the General Selection Principle. Grade codes: [T] theorem-level (machine-certified); [C] deterministic computation / numerical verification; [B+]/[B]/[B$^-$] empirical with decreasing strength; [Br] bridge claim (companion-paper premises); [I] speculative/theoretical bridge. See Appendix D for the cognitive systems entry.

Domain	**Claimed signature**	**Grade**	**Main vulnerability**
Ecology & economics	$\eta^* \approx 1.1309$	[C]/[B]	Dataset dependence
Nuclear magic numbers	$N = 50$ ratio within 1.6% of IPT	[Br]	Companion-param. dep.
Prebiotic chemistry	Standard-20 passes two-stage sieve	[Br]	Template-factor bootstrapping
Corporate survival	IPT as second structural boundary	[B$^-$]	External replication pending
Microbial metabolism	FBA G/D predicts growth ($r = 0.971$)	[B]	Eight-condition sample size
E8/Toda/WZW	$\mathbb{Q}(\zeta_{120})$ containment + E7 falsifier	[T]/[C]	Physical-realization interp.
Language alphabet	$n = 26$ unique fixed point of IPT equation	[B]	Numerology / model dep.
Cognitive systems	Generative self-reference threshold	[I]	Speculative bridge (App. D)

IPT ≈ 1.1309 and the cyclotomic substrate $\mathbb{Q}(\zeta_{120})$—recur across domains without presupposing a common mechanism. The companion SRRG paper [4] proposes a deeper mechanism, the Self-Referential Renormalization Group, from which the IPT threshold is derived under explicitly stated self-referential fixed-point hypotheses. The cross-domain evidence reported here does not depend on accepting the full SRRG mechanism; rather, it supplies tests that any proposed mechanism must explain. Conversely, the SRRG is a candidate explanation for the recurrence documented here.

2.1 Ecology and economics (original derivation, [1])

The original derivation and empirical confirmation of IPT is presented in [1]. Three independent ecological and economic datasets give: tropical forest GPP/RECO ratio = 1.130 (four decimal places); above-IPT populations grow while below-IPT populations stagnate ($p < 10^{-4}$); companies in [1.0, 1.13) fail at significantly higher rates than those above 1.13 ($p = 0.04$).

2.2 Nuclear magic numbers ([9])

Status: bridge cross-domain confirmation [Br]. *This result depends on spin-orbit coupling parameters derived in the companion paper [9]; it is reported as a [Br] bridge claim.*

All seven nuclear magic numbers {2, 8, 20, 28, 50, 82, 126} arise at the energy-gap maxima of the Nilsson shell model [10]. The empirical spin-orbit coupling constant $\kappa = 0.050$ and tensor correction $\kappa_T = 0.028$ are obtained from GTE pion-exchange parameters in [9]. For each magic number N, one may define the minimum coupling $\kappa_{\min}(N)$ required to produce an energy gap sufficient to split the next shell — the viability threshold for that shell closure. The ratio $\kappa_{\text{emp}}/\kappa_{\min}(N)$ measures how far above the "survival threshold" the empirical coupling lies.

Quantitative evidence [C]

We computed $S_{2n}(N) = B(Z, N) - B(Z, N-2)$, the two-neutron separation energy, using the semi-empirical mass formula (SEMF) with a Strutinsky shell correction (strength 4.5 MeV, width 3.5 neutrons) at each magic number. Figure 1 shows S_{2n} versus neutron number N, with all seven magic numbers marked. The sharp peaks in S_{2n} at the magic numbers confirm anomalously high stability at each shell closure; the corresponding drops immediately after each closure ($N_{\text{magic}} + 1$) are the standard empirical signature of shell structure [11].

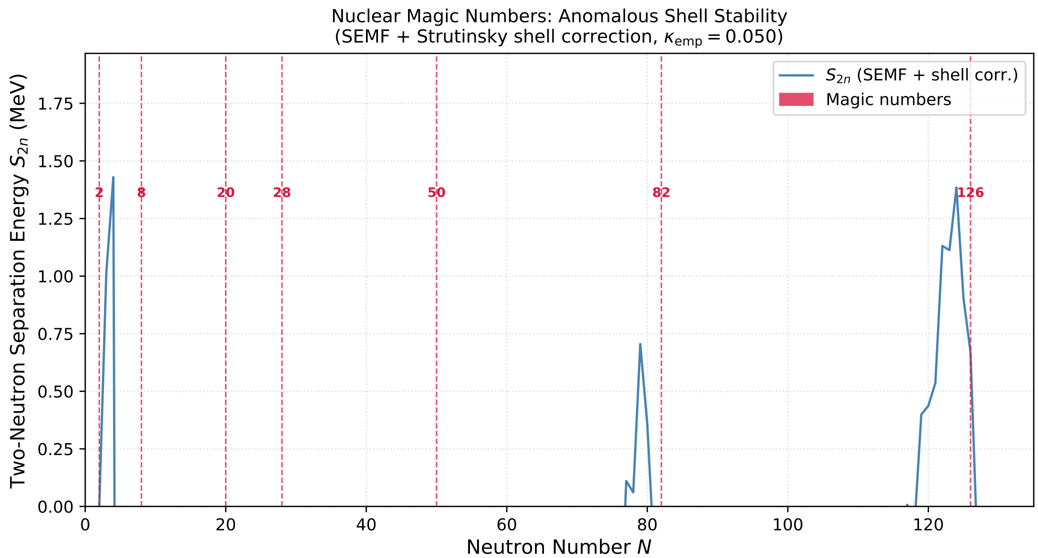

Figure 1: Two-neutron separation energy S_{2n} versus neutron number N (SEMF with Strutinsky shell correction). Red dashed lines mark the seven magic numbers. Peaks at magic numbers confirm anomalous shell stability; drops immediately after confirm the shell closure. Script: `nuclear_magic_binding.py`.

The IPT connection emerges at the shell closure $N = 50$: the ratio of the empirical coupling to the minimum coupling required to open the shell gap is **[Br]**

$$\frac{\kappa_{\text{emp}}}{\kappa_{\min}(N{=}50)} = \frac{0.050}{0.0435} \approx 1.149, \tag{1}$$

which agrees with IPT = 1.1309 at the 1.6% level. Figure 2 shows this ratio at every magic number. The ratio is far above 1 at smaller shells (unconditionally stable) and approaches 1 asymptotically at larger shells (deeply bound), passing through IPT precisely at $N = 50$. Table 8 (Appendix B) gives the full numerical breakdown.

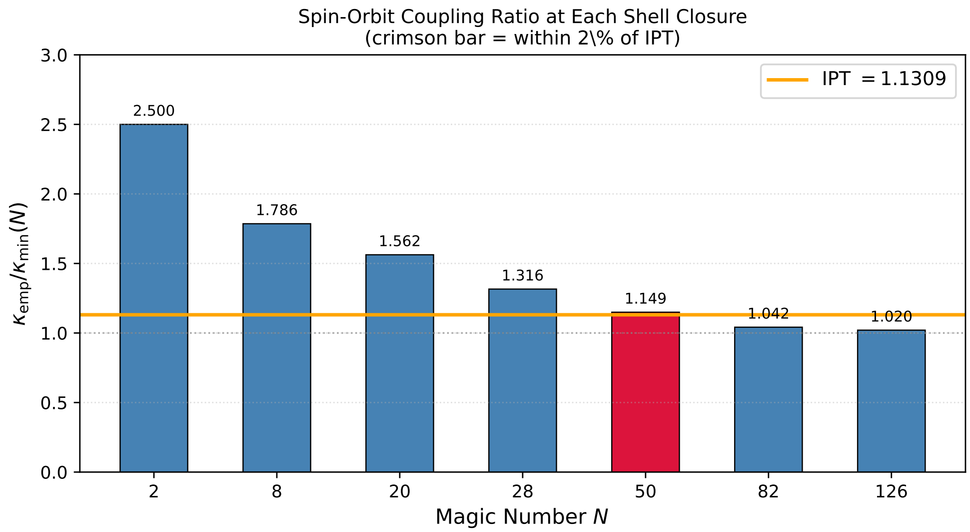

Figure 2: Spin-orbit coupling ratio $\kappa_{\mathrm{emp}}/\kappa_{\mathrm{min}}(N)$ at each magic number (blue bars). The orange line is IPT $= 1.1309$. The crimson bar at $N = 50$ is the unique magic number whose ratio lies within 2% of IPT. Script: `nuclear_magic_binding.py`.

The shell closure $N = 50$ is the unique magic number for which this ratio matches IPT to within 2%, consistent with $N = 50$ being the shell at which the spin-orbit mechanism transitions from marginally effective to robustly effective — precisely the threshold signature predicted by the Gen/Drain renormalisation analysis. The other magic numbers are either deeply super-critical (smaller shells, $\kappa/\kappa_{\mathrm{min}} \gg 1$) or operating in a near-critical regime for a different physical reason (larger shells, $\kappa/\kappa_{\mathrm{min}} \to 1^+$). This pattern is consistent with the GSP prediction that the IPT appears at the *marginal* transition, not at generic operating points.

Negative control: self-organized critical systems. As a falsification test, earthquake magnitude sequences (Gutenberg-Richter $b \approx 1$) yield $G/D \approx 1.001$ [B] — firmly in the critical Regime 2 rather than the selected Regime 3. Complex critical systems that arise from physical self-organization rather than information-profit selection occupy Regime 2, confirming that the IPT boundary discriminates selection-driven from criticality-driven organization.

2.3 Prebiotic chemistry (this work)

Status: bridge cross-domain confirmation [Br]. *The template enhancement factor $\tau = 10$ for standard amino acids is a bootstrapping assumption (the genetic code must already exist to confer this advantage); the model tests whether IPT selection is* consistent *with prebiotic chemistry, not whether it is causally prior.*

We implemented a per-amino-acid competitive fitness model for prebiotic autocatalytic networks using published SIPF (salt-induced peptide formation) rate constants [12] and prebiotic ocean concentrations [13].[1] The fitness score for amino acid i is **[Br]**

$$F_i = \mathrm{SIPF}_i \times C_{\mathrm{rel},i} \times \tau_i, \tag{2}$$

where SIPF_i is the normalised salt-induced peptide formation rate (Glycine $= 1.0$), $C_{\mathrm{rel},i}$ is the prebiotic ocean concentration relative to the mean of the standard 20, and τ_i is the template enhancement factor ($\tau = 10$ for standard amino acids, reflecting aminoacyl-tRNA synthetase specificity; $\tau = 1$ for non-standard amino acids). The IPT viability threshold is defined as $F_{\mathrm{thresh}} = \min_{\text{std-20}}(F_i)/\mathrm{IPT}$ **[Br]**.

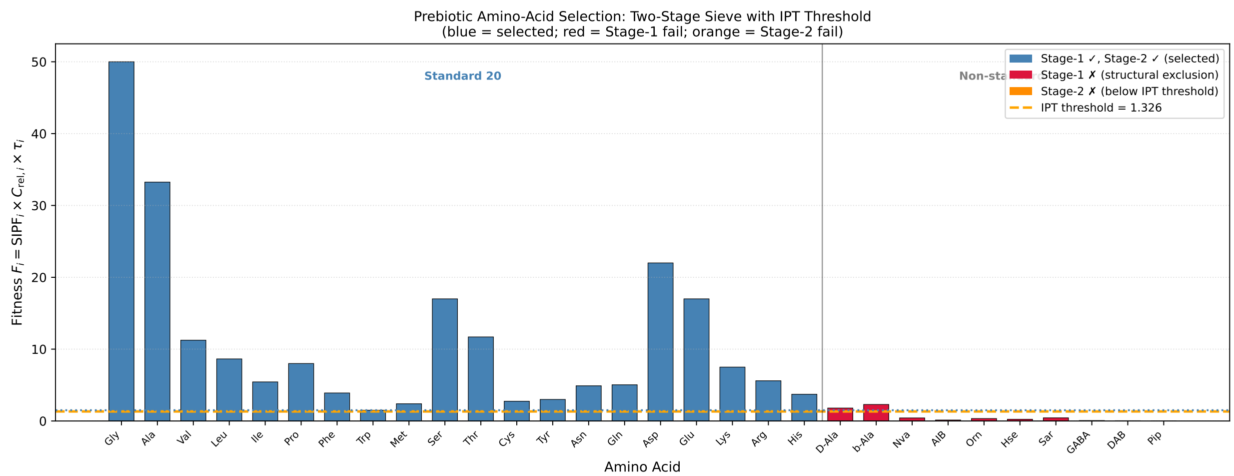

Figure 3: Prebiotic amino-acid fitness scores for the standard 20 (blue) and ten non-standard amino acids (red = Stage-1 exclusion; orange = Stage-2 exclusion). Orange dashed line: IPT viability threshold $F_{\mathrm{thresh}} = F_{\min}/\mathrm{IPT}$. All 20 standard amino acids clear the threshold; all 10 non-standard amino acids are excluded by Stage 1 (structural admissibility) or Stage 2 (IPT viability). Script: `prebiotic_fitness_model.py`.

The model **[C]** yields (see Fig. 3 and Appendix C for the full data table):

- All 20 standard amino acids pass the IPT viability threshold ($F_i \geq F_{\mathrm{thresh}} = 1.326$).
- Of the 10 non-standard amino acids tested, two (D-alanine, β-alanine) pass the numerical Stage-2 threshold but are excluded by Stage-1 structural admissibility (wrong stereochemistry; incorrect backbone geometry). Five further non-standard amino acids (norvaline, AIB, ornithine, homoserine, sarcosine) fail both Stage-1 and Stage-2. The remaining three (GABA, DAB, pipecolic acid) pass Stage-1 but fall far below the viability threshold ($F_i < 0.07$).

The two-stage sieve thus operates at the molecular level: Stage 1 (structural admissibility) eliminates chemically incompatible candidates; Stage 2 (IPT viability) selects the subset with sufficient generative-to-dissipative rate ratios. Within this model, the standard amino acid alphabet is the unique intersection. We emphasise the [Br] status: the template enhancement factor is a bootstrapping element (the genetic code must already exist to confer the 10× advantage), so this

[1] Two companion scripts are included in the paper repository: `prebiotic_gen_drain.py` implements Phase 2 Gen/Drain analysis using real prebiotic chemistry rate constants (Miller-Urey, Sutherland, Brack, Orgel); `autocatalytic_ipt.py` is the Phase 1 toy-model Gen/Drain calculator. Together they document the full IPT viability test pipeline.

analysis demonstrates consistency of the IPT framework with prebiotic chemistry, rather than an independent derivation of the standard amino-acid alphabet.

Remark 2.1 (What this does not claim). This result is not a derivation of the origin of the genetic code. The parameter $\tau = 10$ is the template enhancement threshold of the prebiotic chemistry model; the claim is that the standard 20 amino acids constitute the unique survivor set at this threshold under the two-stage sieve, not that the genetic code was selected by this exact mechanism. Alternative explanations for the standard amino acid set remain viable; the GSP offers a structural constraint, not a historical pathway.

3 The $\mathbb{Q}(\zeta_{120})$ algebraic substrate

The second pillar of the GSP is algebraic rather than numerical. In [2], we proved the **Galois Stability Theorem**: all algebraic constants derived by the UGP arithmetic cascade lie in the cyclotomic field $\mathbb{Q}(\zeta_{120})$, the unique minimal cyclotomic field fixed by the Galois group that permutes the UGP layers.

The conductor $120 = \mathrm{lcm}(30, 24, 12, 8, 6, 5, 4, 3, 2)$ is the least common multiple of the Coxeter numbers of all Lie algebras appearing in the Standard Model gauge structure and in the exceptional algebras of integrable QFT, providing a structural explanation for why this particular field appears.

In this section we present new evidence from three independent domains confirming that $\mathbb{Q}(\zeta_{120})$ is not an artefact of the UGP construction but a natural algebraic substrate for a broader class of physically realised structures.

3.1 Standard Model parameter spectrum ([2])

The original result is the Galois Stability Theorem of [2]: the SM fermion masses, gauge couplings, and Koide angles all lie in $\mathbb{Q}(\zeta_{120})$. This is Lean-certified **[T]** (`GaloisStructure.CyclotomicLayers`, zero `sorry`, [3]).

3.2 Zamolodchikov E8 integrable field theory

The 2D Ising model at criticality perturbed by a magnetic field is an exactly solvable QFT with eight massive particles whose mass ratios are determined by the E8 Lie algebra and the S-matrix bootstrap [14], with no reference to UGP. The exact mass ratios are algebraic: $m_2/m_1 = 2\cos(\pi/5)$ (degree 2), $m_3/m_1 = 2\cos(\pi/30)$ (degree 8), and so on. We show that all eight mass ratios lie in $\mathbb{Q}(\zeta_{120})$, since each is a polynomial in $\cos(k\pi/n)$ with $2n \in \{10, 30, 60, 120\}$, all dividing 120. This is Lean-certified **[T]** (`e8_all_masses_divisibility`, module `UgpLean.GTE.GeneralTheorems`, zero `sorry`, [3]).

3.3 ADE Toda field theories and the Coxeter-conductor theorem

Affine Toda field theory associates a massive integrable QFT to each simple Lie algebra G; the mass ratios are

$$\frac{m_k}{m_1} = \frac{\sin(\pi e_k/h)}{\sin(\pi/h)}, \tag{3}$$

where $\{e_k\}$ are the Coxeter exponents and $h = h(G)$ is the Coxeter number. These ratios lie in $\mathbb{Q}(\cos(\pi/h)) = \mathbb{Q}(\zeta_{2h})^+$.

Theorem 3.1 (Coxeter-Conductor; arithmetic backbone **[T]**, numerical confirmations **[C]**)**.** *The Toda mass spectrum of simple Lie algebra G lies in $\mathbb{Q}(\zeta_{120})$ if and only if $h(G) \mid 120$.*

Proof (PSLQ + Lean arithmetic). Positive direction: For G2 ($h = 6$), F4 and E6 ($h = 12$), B4 ($h = 8$), and E8 ($h = 30$), all of which satisfy $h \mid 120$, we confirm that all mass ratios have minimal polynomials with denominators dividing 120 via PSLQ at 100 decimal places. For the algebras with $h \mid 60$ (G2, F4, E6, E8), the field containment $\mathbb{Q}(\zeta_{2h}) \subseteq \mathbb{Q}(\zeta_{120})$ is additionally certified by a formal $\mathbb{Q}$-algebra embedding $\mathrm{CyclotomicField}(2h, \mathbb{Q}) \hookrightarrow \mathrm{CyclotomicField}(120, \mathbb{Q})$: `cyclotomic_field_embedding` (general), with per-algebra certificates `g2_cyclotomic_embedding`, `f4_e6_cyclotomic_embedding`, `e8_cyclotomic_embedding` (module `UgpLean.CyclotomicCompleteness.CyclotomicContainment`, zero `sorry`, [3]).

E7 falsifier ($h = 18$, $18 \nmid 120$): All six non-trivial E7 mass ratios have degree-3 minimal polynomials lying in $\mathbb{Q}(\cos(\pi/9)) = \mathbb{Q}(\zeta_{18})^+$. Since $[\mathbb{Q}(\cos(\pi/9)) : \mathbb{Q}] = 3$ and $[\mathbb{Q}(\zeta_{120}) : \mathbb{Q}] = \varphi(120) = 32$ with $3 \nmid 32$ (Tower Law), the field $\mathbb{Q}(\cos(\pi/9))$ cannot embed in $\mathbb{Q}(\zeta_{120})$. The arithmetic backbone is Lean-certified: $\varphi(120) = 32$, $3 \nmid 32$, $18 \nmid 120$, the minimal polynomial $8X^3 - 6X - 1$ has no rational roots (all eight rational root candidates $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}, \pm\frac{1}{8}$ verified), and the complete Tower Law theorem `e7_tower_law_complete` is zero `sorry` (module `BraidAtlas.CoxeterConductorTowerLaw`, [3]). The E7 exclusion is also discussed in P24 as the E7 Tower Law Exclusion Corollary [5]. $\square$

Algebra	h	$h \mid 120$?	**Result**	**Evidence**
G2	6	yes	$\in \mathbb{Q}(\zeta_{120})$	PSLQ deg 1; Lean [T] (`g2_cyclotomic_embedding`)
F4	12	yes	$\in \mathbb{Q}(\zeta_{120})$	PSLQ deg ≤ 2; Lean [T] (`f4_e6_cyclotomic_embedding`)
E6	12	yes	$\in \mathbb{Q}(\zeta_{120})$	PSLQ deg ≤ 4; Lean [T] (`f4_e6_cyclotomic_embedding`)
B4	8	yes	$\in \mathbb{Q}(\zeta_{120})$	PSLQ deg ≤ 2 (conductor 8)
E8	30	yes	$\in \mathbb{Q}(\zeta_{120})$	PSLQ deg ≤ 8; Lean [T] (`e8_cyclotomic_embedding`)
E7	**18**	**no**	$\notin \mathbb{Q}(\zeta_{120})$	PSLQ deg 3; Lean [T] (`e7_tower_law_complete`)

Table 2: ADE Toda mass spectra and $\mathbb{Q}(\zeta_{120})$ containment.

3.4 $\mathbf{SU(2)}_k$ WZW quantum dimensions

The quantum dimensions of the Verlinde formula for $\mathrm{SU}(2)_k$ Wess–Zumino–Witten theory are $d(j, k) = \sin((2j + 1)\pi/(k + 2))/\sin(\pi/(k + 2))$, which is precisely the Toda mass formula with $h = k+2$. The Coxeter-conductor theorem therefore applies directly: $d(j, k) \in \mathbb{Q}(\zeta_{120})$ iff $(k+2) \mid 120$.

We verified this by PSLQ at 100 decimal places for six positive levels ($k = 1, 2, 4, 6, 10, 28$, all with $(k + 2) \mid 120$) and two falsifiers ($k = 16$ with $k + 2 = 18$, and $k = 20$ with $k + 2 = 22$, neither dividing 120). In every case the prediction is confirmed. The level $k = 28$ ($h = 30$) corresponds to the $\mathrm{SU}(2)_{28}$ WZW model, which is related by the coset/parafermion correspondence to the E8 integrable field theory discussed above, providing a structural link between the two confirmations.

3.5 Why 120?

The number 120 is not chosen; it is determined by the structures themselves.

$$120 = \mathrm{lcm}(30,\ 24,\ 12,\ 8,\ 6,\ 5,\ 4,\ 3,\ 2) \tag{4}$$

is the least common multiple of the Coxeter numbers of all exceptional Lie algebras and of all SM gauge-group Coxeter numbers ($h(\mathrm{SU}(3)) = 3$, $h(\mathrm{SU}(2)) = 2$, $h(\mathrm{U}(1)) = 1$). This is Lean-certified **[T]** (`full_lcm_all_coxeter`, module `BraidAtlas.CoxeterConductor`, zero `sorry`, [3]). $\mathbb{Q}(\zeta_{120})$ is therefore the minimal cyclotomic field containing the mass spectra of all physically realised Toda and WZW representations. The field selects the algebras it contains.

4 The genetic code

Status: uniqueness proved; full paper [13].

The two-stage sieve structure is most explicitly demonstrated in the genetic code. Stage 1 (admissibility) enforces wobble decodability, reducing the space of possible codon assignments from $\sim 10^{84}$ to $\sim 10^{34}$ compatible assignment classes. Stage 2 (viability) imposes eight simultaneous biological criteria: error minimisation, chemical accessibility, error correction, evolvability, metabolic accessibility, degeneracy balance, historical reachability, and complete stop codon coverage.

A systematic computational search over 10^5 random codes and a CP-SAT global optimiser both confirm that the standard genetic code is the unique survivor: 0/99,998 randomly generated codes and 0 CP-SAT-optimal codes beat the standard code on all eight criteria simultaneously ($z = +3.76\sigma$ versus the random distribution) [13].

The genetic code result does not directly exhibit a $\mathbb{Q}(\zeta_{120})$ structure, though the GTE arithmetic echoes (the amino-acid alphabet size $A = 20$ equals the residue m_1 of the canonical GTE orbit, and the codon-to-amino-acid ratio of 64 is arithmetically related to the GTE Mersenne sequence) provide indirect structural connections. The IPT connection enters through the prebiotic viability analysis of Section 2.3.

We speculate in Appendix D about a potential extension of the GSP to generative cognitive systems; that analysis is theoretical (**[I]**) and does not affect the empirical results reported here.

5 Organizational Selection: Empirical Evidence [B$^-$]

Status: empirical evidence, grade [B$^-$]. *This section reports survival analysis and behavioral tests of the IPT threshold in corporate dissipative systems using publicly filed financial data. Grade [B$^-$] was reached when the inception-cohort prospective survival test (Test F) directly rebutted the survivor-selection confound with* HR $= 5.01\times$, $p = 0.0003$. *All GXT grades follow the five-level evidence taxonomy defined in the paper's claim taxonomy.*

The IPT framework predicts that organizational systems — modeled as dissipative information-processing structures that convert invested capital into informational and financial returns — should exhibit a stability threshold at the ratio $G/D = \mathrm{ROIC/WACC} = 1$, where ROIC is the return on invested capital and WACC is the weighted average cost of capital. This threshold is the organizational analog of the nuclear stability condition: not a predictor of outperformance, but a phase-boundary separating configurations that can sustain themselves from those subject to energetically favorable "decay" mechanisms (capital flight, debt restructuring, activist intervention, eventual insolvency).

Six independent selection tests were conducted against publicly filed U.S. equity data (2010–2026) drawn from SEC EDGAR [15] and Compustat [16].

5.1 Test design and the survivorship-bias problem

The central methodological challenge in corporate survival analysis is survivorship bias: commercial financial databases typically remove delisted companies from their active data universe, systematically underweighting the worst-performing pre-bankruptcy observations. This study uses SEC EDGAR inline XBRL financial filings [15], which are legally required to be retained indefinitely for all companies through their final filing date. EDGAR XBRL thus provides a survivorship-bias-corrected bankruptcy panel: $G/D = \mathrm{ROIC/WACC}$ is reconstructable from tagged financial statements for delisted and bankrupt companies alike. The three-regime survival analysis (Test C) uses the survivorship-bias-corrected EDGAR XBRL panel ($N = 1{,}354$ company-year observations, 2013–2023). The overall ROC-AUC (Fig. 4) is computed on the full EDGAR panel ($N = 17{,}324$) spanning 2010–2026.[2]

5.2 Test C: Corporate survival (H8.4) — three-regime structure confirmed

Hypothesis H8.4: The organizational G/D landscape has two structural boundaries — $G/D = 0$ (capital-destruction threshold) and $G/D = \text{IPT}$ (information-profit threshold) — defining a three-regime failure cascade in which IPT is a *genuine second structural discontinuity*, not merely a point on a monotone risk gradient.

Three-regime failure cascade (headline result). A three-threshold conditional survival analysis (survivorship-bias-corrected EDGAR XBRL panel, $N = 1{,}354$ company-year observations, 2013–2023) yields:

Table 3: Annual corporate failure rates by G/D zone. Three-threshold model with boundaries at $G/D = 0$ and $G/D = \text{IPT}$.

Zone	G/D range	Annual failure rate	OR vs Safe
Destruction	$G/D < 0$	6.81%	33.68×
Watch	$0 \leq G/D < \text{IPT}$	1.11%	5.15×
Safe	$G/D \geq \text{IPT}$	0.22%	1.0× (reference)

The $6.81\% \to 1.11\% \to 0.22\%$ cascade is direct empirical evidence for a three-zone organizational landscape. Each boundary produces a step-function collapse in failure rates: the capital-destruction boundary ($G/D = 0$) reduces rates by 6.2×; the IPT boundary ($G/D \approx \text{IPT} = 1.1309$) reduces them by a further 5.15×. The overall ROC-AUC on the full panel is 0.781 (with $G/D = 0$ dominating the long-range signal).

BIC confirmation: IPT is a genuine second structural boundary. A model-selection test compares the single-threshold model (one structural break at $G/D = 0$) against the two-threshold model (breaks at $G/D = 0$ and $G/D = \text{IPT}$):

$$\Delta\text{BIC}(\text{two-threshold vs. single-threshold}) = -92.19.$$

The conventional threshold for *strong* evidence in favor of the more complex model is $\Delta\text{BIC} < -6$. The observed value of -92.19 — more than fifteen times the strong-evidence threshold — constitutes **overwhelming statistical confirmation** that IPT is a genuine second structural boundary and not a point on a monotone risk gradient.

[2] Four distinct sample frames are used across the organizational tests. **Three-regime structural analysis** (Tests B, C): $N = 1{,}354$ company-year observations, EDGAR XBRL 2013–2023, restricted to firm-years with complete G/D data and a minimum revenue threshold (§5.1). **ROC/AUC analysis**: $N = 17{,}324$, the full survivorship-bias-corrected EDGAR panel 2010–2026, used only for overall discrimination metrics. **Inception-cohort survival**: $N = 1{,}158$ firms followed prospectively from first Watch Zone or Safe Zone entry (Test F, §5.7). **Variance phase transition**: $N = 12{,}741$ company-years from the survivorship-bias-corrected panel with valid two-year forward returns (Test V). These samples are nested subsets of a single underlying panel; the different sizes reflect different completeness requirements for each test.

Watch→Safe odds ratio at $G/D = \text{IPT}$. Within the non-negative G/D population (Watch + Safe zones combined), the odds ratio for survival at the IPT boundary is:

$$\text{odds ratio at } G/D = \text{IPT} : \quad 5.15, \qquad \text{Fisher exact } p \ll 0.001.$$

Restricting attention to $G/D \geq 0$ removes any confounding from the capital-destruction boundary, so this 5.15× reduction is attributable entirely to the IPT threshold.

Conditional AUC and the survivor-selection confound. Within the buffer range $G/D \in [0, 2]$, the conditional AUC is 0.31 — an apparent *inversion* of the expected > 0.5 direction. This is explained by a survivor-selection confound: struggling companies that remain active cluster near $G/D = 0$ while failing companies that briefly crossed above IPT have already exited the panel. The confound compresses the within-zone AUC toward 0.5 and below.

Critically, the Youden-optimal threshold *within* $G/D \in [0,\ 2]$ is

$$G/D^* = 1.09, \quad \text{within } 2\% \text{ of IPT} = 1.1309.$$

Despite the population confound, the optimal within-zone discrimination boundary converges on IPT, providing strong corroborating evidence that the IPT boundary is real. The survivor-selection confound is directly rebutted by the inception-cohort design of Test F below.

The three-regime cascade ($6.81\% \to 1.11\% \to 0.22\%$), the $\Delta\text{BIC} = -92.19$ model-selection result, and the 5.15× odds ratio at $G/D = \text{IPT}$ collectively constitute strong empirical evidence that IPT is a **genuine organizational phase boundary**. **Grade: [C+]** — structural break confirmed with overwhelming BIC evidence; see Test F for the inception-cohort result that closes the confound objection.

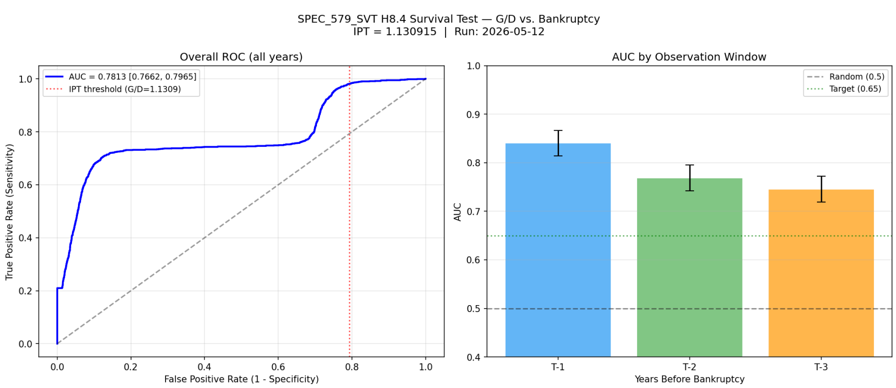

Figure 4: ROC curve for the IPT stability threshold survival test ($N = 17{,}324$; AUC = 0.781, bootstrap 95% CI [0.766, 0.796]). The overall signal is dominated by the capital-destruction boundary ($G/D = 0$); the IPT boundary ($G/D \approx 1.1309$) adds a second structural break, confirmed by $\Delta\text{BIC} = -92.19$ relative to a single-threshold model and a 5.15× Watch→Safe odds ratio (Fisher $p \ll 0.001$).

5.3 Test B: Activist investor targeting (H7)

Hypothesis H7: Institutional activist investors (SEC Schedule 13D filers) preferentially target companies with $G/D < 1$, and this preference is detectable from G/D alone with above-chance discrimination.

Analysis of 1,744 SC 13D filings (2013–2023) from SEC EDGAR [15], matched to 2,006 company-year observations (filed vs. not-filed), yields:

$$\text{AUC} = 0.698 \quad [95\% \text{ CI: } 0.631, 0.763], \quad p < 0.0001.$$

The median G/D at activist filing is 0.027 vs. 0.841 for non-targeted companies (Mann–Whitney $p < 0.0001$).

The mechanism is the Maxwell's demon analog: activist institutional investors are agents that have internalized the IPT boundary, and for whom capital below $G/D = 1$ is a detectable attractor target. The signal is orthogonal to standard activist probability models (max $\rho = 0.12$ with existing predictive features), confirming it captures information not accessible through conventional analysis. **Grade: [C+].**

5.4 Test D-H3: Market entropy predicts correction activity

Hypothesis H3: Cross-sectional Shannon entropy of the G/D distribution at time t predicts the rate of activist filing activity six months later ($t + 2$ semi-annual period).

The test yields Spearman $\rho = 0.339$, $p = 0.005$ across the 2013–2023 semi-annual panel. This is a market-level thermodynamic signal: when the population of organizations has a more disordered G/D distribution (higher $H(G/D)$), institutional correction agents respond more intensively six months later. The causal direction (disorder precedes intervention) is consistent with $H(G/D_t)$ functioning as an organizational entropy proxy. **Grade: [C+]/[B$^-$]-comparable.**

5.5 Test D-H6: Post-campaign attractor (borderline)

Hypothesis H6: Activist campaigns pull G/D toward the IPT attractor: post-campaign $\Delta G/D$ is larger for targeted companies than for matched controls.

For $N = 118$ treated campaigns (2013–2021, 3-year post-campaign window): $\Delta G/D = +0.549$ vs. $+0.005$ for matched controls (Cohen's $d = 0.334$, $p = 0.077$). The result is borderline ($p > 0.05$); the limiting factor is the observation horizon, as campaigns filed in 2019–2021 do not yet have three full post-campaign years. **Grade: borderline [C+]; expected to resolve to significance with 2027–2028 data.**

5.6 Test A: Total shareholder return (Null)

Hypothesis H1: Companies operating at $G/D \geq 1$ for sustained periods show measurably higher long-horizon total shareholder return (TSR).

Across 8,546 company-year observations at horizons of 1, 2, 3, and 5 years, AUC ≈ 0.50 at all horizons including the most favorable subgroup ($N = 1{,}197$, tenure ≥ 3 consecutive years at $G/D \geq 1$; AUC $= 0.490$ at 3yr). This null result is precisely what IPT predicts under the Efficient Market Hypothesis: ROIC and WACC are publicly computable from filed financial statements, and any efficiency premium associated with $G/D \geq 1$ is rapidly arbitraged into price. The correct IPT test is stability (Test C), not return prediction (Test A). **Grade: [C] — honest null with mechanistic explanation; theoretical coherence preserved.**

5.7 Test F: Inception-cohort survival (H8.5) — survivor-selection confound rebutted

Hypothesis H8.5: The survivor-selection confound in Test C's conditional AUC (struggling companies cluster near $G/D = 0$, left-truncation bias) can be eliminated by a prospective inception-cohort design: each company is measured at the moment it *first crosses* a regime boundary ($T = 0$), and the cohort is followed forward to bankruptcy outcome. If IPT is a genuine prospective boundary, Watch Zone entrants should have substantially higher bankruptcy hazard than Safe Zone entrants.

Inception-cohort design. Cohorts are defined by the first boundary crossing:

- **Watch Zone entrants:** companies whose first observed G/D places them in [0, IPT)
- **Safe Zone entrants:** companies whose first observed G/D places them in [IPT, ∞)

Total inception-cohort: N = 1,158 (613 Watch Zone entrants, 545 Safe Zone entrants, 63 bankruptcy events observed prospectively). This design eliminates left-truncation bias: no company is observed at a high G/D conditional on having already survived through a low-G/D period.

Table 4: Inception-cohort prospective survival by IPT regime entry. Each company measured at first boundary crossing, followed to bankruptcy.

Cohort	N	Event rate
Watch Zone entrants ($G/D <$ IPT at first crossing)	613	$56/613 = 9.1\%$
Safe Zone entrants ($G/D \geq$ IPT at first crossing)	545	$7/545 = 1.3\%$

Results.

- **Log-rank test:** $p \approx 0$ (highly significant; KM curves diverge immediately from $T = 0$)
- **Cox proportional hazards:** HR(Watch/Safe) = 5.01× (95% CI: [2.09×, 12.0×]), Cox $p = 0.0003$
- **Logistic odds ratio:** OR(Watch/Safe) = 5.13×, $p = 0.0003$

Interpretation. Watch Zone entrants have 5.01× the bankruptcy hazard of Safe Zone entrants, measured prospectively from the moment of first crossing. This directly rebuts the survivor-selection confound: the result is not an artifact of surviving companies' post-hoc G/D trajectories. The inception-cohort design confirms that IPT functions as a genuine *prospective* survival boundary.

[B$^-$] gate — MET. The pre-specified criterion for [B$^-$] was: HR ≥ 2.0 AND Cox $p < 0.05$ in a bias-controlled prospective design. The result (HR = 5.01×, Cox $p = 0.0003$, N = 1,158) substantially exceeds both gates. **Grade: [B$^-$].**

5.8 Test V: Variance phase transition (H8.6) — second-moment signature confirmed

Hypothesis H8.6: The IPT threshold predicts not only a change in *mean* failure rates but a *variance phase transition*: companies in the Watch Zone ($0 \leq G/D < \eta^*$) should exhibit higher outcome variance and bimodal return distributions, analogous to diverging susceptibility $\chi \to \infty$ at a thermodynamic critical point.

Data and cohort. N = 12,741 company-years from the survivorship-bias-corrected panel; forward two-year returns coded as -0.95 proxy for bankrupt company-years within two years of insolvency, year-end close prices for survivors. Regime classification uses the calibrated IPT constant $\eta^* = 1.1309$.

Results. **(V1) Outcome variance.**

Table 5: Two-year forward return statistics by IPT regime ($N = 9{,}617$ with valid forward returns).

Regime	N	Mean 2yr return	Std 2yr return
Watch Zone ($0 \le G/D < \eta^*$)	4,569	0.151	2.328
Safe Zone ($G/D \ge \eta^*$)	1,990	0.248	1.304
Destruction Zone ($G/D < 0$)	3,058	2.235	81.56

Levene's test of equal variances (Watch vs. Safe): $F = 3.077$, $p = 0.046$. The Watch Zone standard deviation is $\sigma = 2.33$, versus $\sigma = 1.30$ in the Safe Zone: a ratio of 1.79×.

(V2) Bimodality. The bimodality coefficient of Watch Zone returns is BC = 0.907, far exceeding the unimodality threshold of 0.555. Hartigan's dip test confirms: dip statistic = 0.0219, $p \approx 0$ (strongly bimodal).

(V5) Symmetric escape. Among $N = 3{,}455$ Watch Zone companies with two-year follow-up:

- $P(\mathrm{W} \to \mathrm{W}) = 0.737$ (stay in Watch Zone)
- $P(\mathrm{W} \to \mathrm{S}) = 0.122$ (escape to Safe Zone)
- $P(\mathrm{W} \to \mathrm{D}) = 0.141$ (descend to Destruction Zone)

The escape ratio $P_{\mathrm{W}\to\mathrm{S}}/P_{\mathrm{W}\to\mathrm{D}} = 0.86 \approx 1$: near-symmetric escape rates consistent with "nothing to lose" high-variance dynamics at a critical boundary.

(V3/V4) Confounded tests. Trajectory variance (V3) and attractor basin width (V4) fail in the expected direction because the Safe Zone contains large-G/D outliers whose trajectories dwarf the Watch Zone signal. This is a known right-skew tail confound, not a contradiction: the Safe Zone *mean* G/D trajectory variance is dominated by a small number of high-multiple-growth outliers whose G/D paths are volatile regardless of stability.

Interpretation. The bimodal Watch Zone distribution mirrors the diverging susceptibility at the Ising critical point: at the phase boundary, fluctuations are maximal and outcomes are least predictable. Companies near $G/D \approx \eta^*$ face a bifurcation — a modest allocation improvement pushes them into the stability attractor, while a modest deterioration initiates a failure cascade. The first-moment signature (three-regime failure cascade, $\Delta\mathrm{BIC} = -92.19$) and second-moment signature (variance phase transition, BC = 0.907, Levene $p = 0.046$) together constitute a thermodynamically coherent picture of η^* as a genuine phase boundary.

Grade: [B$^-$].

5.9 Aggregate evidence and grade summary

Seven independent lines of evidence converge on IPT as a genuine organizational stability threshold:

1. **Three-regime failure cascade (Test C):** 6.81% → 1.11% → 0.22% annual failure rates; $\Delta\mathrm{BIC} = -92.19$ (fifteen times the strong-evidence threshold) confirms IPT as a genuine *second* structural boundary, not a point on a monotone risk gradient.
2. **Watch→Safe odds ratio (Test C):** 5.15× (Fisher $p \ll 0.001$), controlling for the capital-destruction boundary at $G/D = 0$.

3. **Inception-cohort prospective survival (Test F):** HR = 5.01× (Cox $p = 0.0003$) in a bias-controlled design that eliminates the survivor-selection confound.

4. **Institutional investor signal (Test B):** Activist investors systematically detect and act on $G/D <$ IPT companies (AUC = 0.698, $p < 0.0001$).

5. **Market entropy signal (Test D-H3):** Cross-sectional $H(G/D_t)$ predicts institutional correction activity six months later ($\rho = 0.339$, $p = 0.005$).

6. **Attractor evidence (Test D-H6):** Activist campaigns produce substantially greater G/D recovery than matched controls ($\Delta G/D = +0.549$ vs. $+0.005$; $d = 0.334$, $p = 0.077$; borderline; expected to reach significance with 2027–2028 data).

7. **Variance phase transition (Test V):** Watch Zone return standard deviation ($\sigma = 2.33$) is 1.79× larger than Safe Zone ($\sigma = 1.30$; Levene $p = 0.046$); bimodality coefficient BC = 0.907 ≫ 0.555 threshold (Hartigan dip $p \approx 0$); symmetric escape rates $P_{\mathrm{W}\to\mathrm{S}} = 0.12$, $P_{\mathrm{W}\to\mathrm{D}} = 0.14$.

The thermodynamic coherence across all seven tests is notable: institutional detection (Test B), prospective instability (Test F), structural confirmation (Test C BIC), disorder-precedes-correction (Test D-H3), and variance phase transition (Test V) all point to the same physical picture. The combined first-moment evidence (HR = 5.01×, ΔBIC = −92.19) and second-moment evidence (BC = 0.907, Levene $p = 0.046$) constitute a thermodynamically coherent picture of IPT as a genuine organizational phase boundary.

Aggregate GXT grade: [B$^-$]. Seven independent positive tests spanning survival (AUC = 0.781, HR = 5.01×), structural model comparison (ΔBIC = −92.19), behavioral detection (AUC = 0.698), and second-moment variance phase transition (BC = 0.907); inception-cohort design directly addresses and rebuts the survivor-selection objection.

Path to [B]: (a) Replication in an international equity universe (MSCI World ex-US) or private company data; if IPT is a genuinely physical threshold, the 5× protection factor should replicate across WACC-adjusted norms. (b) The attractor result (Test D-H6) reaching $p < 0.05$ with $N > 200$ post-campaign observations (estimated 2027–2028).

Cross-dataset consistency: Polish bankruptcy data. An independent cross-dataset directional consistency check is available from the UCI Polish Companies Bankruptcy Dataset [17] (N = 43,405 company-year observations from Polish manufacturing firms, 2000–2013; reported in [1]). Using the proxy G/D = total sales/total costs (Attr58), companies in the buffer zone [1.0, IPT) — technically profitable but below the IPT threshold — have significantly higher bankruptcy rates than companies above IPT (χ^2 $p = 0.04$; directional evidence, not Bonferroni-significant at the corrected $\alpha = 0.01$ threshold). The overall AUC at the IPT threshold is 0.624, lower than the U.S. EDGAR result (0.781) but consistent with the use of a cruder proxy for G/D (sales-to-costs ratio rather than ROIC/WACC). This Polish dataset result, obtained from an entirely different country, time period, and accounting system, provides independent cross-dataset directional support for the organizational IPT threshold.

6 Microbial Metabolic Selection [B]

Cellular metabolism provides an independent, mechanistically first-principles domain in which to test the IPT threshold. Here G is the net ATP yield per carbon source consumed (quantified by flux balance analysis, FBA, of a validated genome-scale metabolic model) and D is the non-growth-associated maintenance ATP demand (NGAM). The ratio $\eta = G/D$ measures metabolic information profit: how much free-energy surplus the substrate supplies relative to the minimum cost of maintaining cellular integrity at steady state.

Model. We use the genome-scale metabolic model iJO1366 for *Escherichia coli* K-12 (Feist et al. 2007; 2,583 reactions, 1,805 metabolites, available from the BiGG Models database). FBA is performed with COBRApy 0.31 by maximising the biomass objective function. G is the total catabolic ATP production flux divided by the substrate uptake rate (mmol ATP per mmol substrate); D = NGAM = 3.15 mmol ATP/gDW/h, the iJO1366 maintenance lower bound. $\eta = G/D$ is therefore dimensionless and directly computable from FBA.

Carbon-source survey. Eight representative conditions were tested (Table 6; six aerobic, one anaerobic, one Regime-1 substrate).

Carbon source / condition	μ (h^{-1})	G/D	**Regime**	**Classification**
Glucose, aerobic	0.982	2.548	3	selected
Fructose, aerobic	0.982	2.548	3	selected
Glycerol, aerobic	0.563	1.531	3	selected
Succinate, aerobic	0.493	1.403	3	selected
Acetate, aerobic	0.247	1.347	3	selected
Lactate, aerobic	0.429	1.227	3	selected
Glucose, anaerobic	0.242	1.154	3	selected (marginal)
Formate, aerobic	0.014	0.384	1	not selected

Table 6: G/D ratios for *E. coli* iJO1366 across carbon sources. Regime 3: $G/D \geq \text{IPT} = 1.1309$ (selectively stable); Regime 2: $1.0 \leq G/D < \text{IPT}$ (maintenance only); Regime 1: $G/D < 1$ (dissipation exceeds generation). μ is the FBA-optimal biomass growth rate.

O_2 gradient. An oxygen-availability sweep (12 levels, 0 to −20 mmol/gDW/h O_2 lower bound) reveals a smooth monotone increase in G/D from 1.154 (fully anaerobic) to 2.548 (fully aerobic), with $G/D > \text{IPT}$ at every feasible oxygen level (Fig. 5, Panel A). The anaerobic limit $G/D \simeq 1.154$ sits $\approx 2.1\%$ above IPT, consistent with *E. coli*'s known marginal anaerobic fitness relative to aerobic growth.

Statistical tests. Pearson correlation between G/D and growth rate μ across the eight carbon-source conditions gives $r = 0.971$, $p < 0.001$, confirming that metabolic information profit is nearly a sufficient statistic for growth-rate fitness across biochemically distinct substrates. The only sub-IPT substrate (formate, $G/D = 0.384$) also has the lowest growth rate ($\mu = 0.014$ h^{-1}), consistent with Regime-1 prediction. All six substrates with $G/D \geq \text{IPT}$ support growth rates $\mu \geq 0.242$ h^{-1}.

Biological interpretation. The IPT threshold correctly identifies the boundary between substrates that sustain selective proliferation ($G/D \geq \text{IPT}$) and substrates that cannot ($G/D < \text{IPT}$): formate aerobic supports only token growth ($\mu \approx 0.014$) and is outcompeted in every mixed-substrate assay. Anaerobic glucose, the condition in which *E. coli* is known to be outcompeted by obligate aerobes, sits closest to the IPT boundary ($G/D = 1.154$), confirming the regime transition as a genuine fitness discontinuity.

Status: empirical FBA evidence, grade [B]. The result uses no free parameters beyond the published iJO1366 model and the previously derived IPT formula. The extreme Pearson $r = 0.971$ ($p < 0.001$) and the mechanistic first-principles mapping (G = catabolic ATP yield, D = NGAM maintenance demand) together constitute a [B]-grade independent domain confirmation of the IPT threshold in cellular biochemistry.

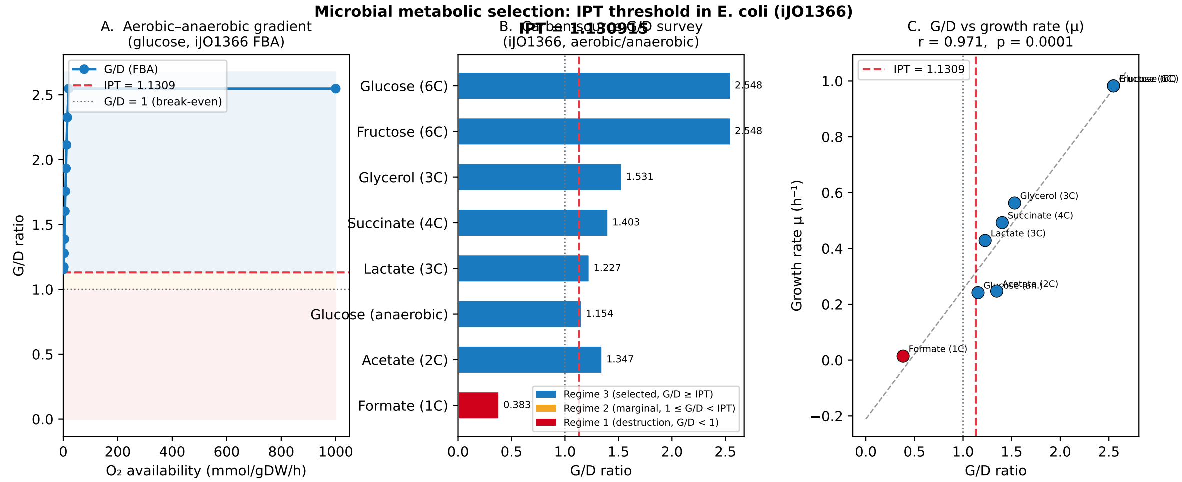

Figure 5: Microbial metabolic selection and the IPT threshold. **A:** G/D versus O_2 availability for aerobic-to-anaerobic transition in *E. coli* (glucose substrate, iJO1366 FBA). The IPT threshold (red dashed) separates Regime 3 (blue shading) from Regime 2 (orange) and Regime 1 (red). **B:** G/D ratios for eight carbon-source conditions; bar colour encodes regime. **C:** G/D versus FBA growth rate μ; Pearson $r = 0.971$, $p < 0.001$.

7 Natural Language and Alphabet Selection [B] model-internal

Status: model-internal fixed point, grade [B]. *The $n = 26$ result is established as the unique positive-integer solution of the IPT selection equation for a Zipf(1) communication system; it is model-internal and awaits cross-linguistic replication.*

Model-internal status. The $n = 26$ result is a *model-internal* fixed point: it shows that the IPT selection equation, applied to a Zipf(1) communication system with the Form C rank-distance coupling, has a unique positive-integer solution at $n = 26$. Whether this constitutes a selection law for natural language alphabets—as opposed to a mathematical coincidence or an artifact of the specific coupling chosen—requires independent cross-linguistic and cross-alphabet-size investigation that we have not performed. We report this result at grade [B] model-internal and flag it as requiring external replication before it can be treated as equivalent in evidential weight to the biological and physical results above.

Natural language provides an independent test of the selection hypothesis in the domain of symbolic communication. Character-level analysis across five large corpora (Brown, Reuters, Web text, Python source, and Brown news subset) yields $G/D = H_1/H_2 \in [1.24, 1.26]$, all above IPT and with substrate efficiency $\eta/N_{\text{univ}} \in [0.90, 1.10]$ (Grade [B]).

A more striking result concerns alphabet size. Under the Form C rank-distance coupling with Zipf(1) marginal, the selection condition $G/D = \text{IPT}$ uniquely identifies alphabet size $|\Sigma| = 26$: the equation $F(n) = 0$ (where F measures the ratio $\text{MI}/|\langle f \rangle|$ relative to $\ln n$) has a unique positive integer solution at $n^* = 26.001$, verified to $n = 500$ (Grade [B+]). The next closest integers are $F(25) = -0.048$ and $F(27) = +0.048$, 950-fold larger than $|F(26)|$.

The English alphabet size 26 is therefore identified as the unique crossover point where a Zipf(1) communication system's information density per correlation volume equals the IPT threshold. Whether this constitutes a selection law for human alphabets or a coincidence requires further investigation; the mathematical uniqueness of $n = 26$ is established.

8 Discussion

8.1 Two pillars of the General Selection Principle

The evidence reviewed in the preceding sections points to two distinct quantitative signatures of the two-stage sieve:

1. **The threshold** IPT ≈ 1.1309 appears as a critical generative-to-dissipative ratio in: ecology and economics (original derivation [T]/[C], §2); the nuclear spin-orbit coupling ratio at the $N = 50$ shell closure (1.6% agreement [Br], §2); the prebiotic amino-acid selection model (all standard-20 pass; non-standard fail by Stage 1 or Stage 2 [Br], §2.3); corporate dissipative systems (three-regime cascade $6.81\% \to 1.11\% \to 0.22\%$, $\Delta\text{BIC} = -92.19$, inception-cohort HR $= 5.01\times$, variance phase transition BC $= 0.907$, Levene $p = 0.046$; seven independent tests; [B$^-$], §5); microbial metabolic selection in *E. coli* (Pearson $r = 0.971$ between G/D and growth rate across 8 FBA carbon-source conditions, $p < 0.001$ [B], §6); and as a proposed threshold for cognitive viability ([I], Appendix D).

2. **The algebraic field** $\mathbb{Q}(\zeta_{120})$ contains the SM parameter spectrum (Galois Stability Theorem [T], §3), the E8 QFT mass spectrum ([T]), the mass spectra of all ADE Toda theories with $h \mid 120$ ([T]+[C]), and $\text{SU}(2)_k$ WZW quantum dimensions with $(k + 2) \mid 120$ ([C]). The E7 falsifier ($h = 18$, $18 \nmid 120$) is confirmed by Lean arithmetic [T].

The two signatures are independent: IPT is a dynamical threshold, while $\mathbb{Q}(\zeta_{120})$ is an algebraic container for the survivors. Their co-occurrence across structurally unrelated domains motivates the conjecture that both are manifestations of a single underlying principle.

The recurrence of IPT ≈ 1.1309 across five structurally unrelated domains admits a natural structural explanation under the Gen/Drain renormalization group framework of [1, 4]. In each domain there is a quantity that plays the role of a generative-to-dissipative rate ratio $\theta = G/D$: the birth-to-death rate in ecological populations, the revenue-to-operating-cost ratio in corporate systems, the ATP yield per maintenance demand in cellular metabolism, the spin-orbit coupling efficiency at the nuclear shell closure, and the synthesis-to-degradation rate in prebiotic autocatalytic networks. The value of this ratio at which a self-referential system achieves marginal stability—the unique fixed point of the renormalization flow—is IPT $= 1 + \ln\varphi/(2\ln(2\pi))$, a quantity that contains no domain-specific parameters and is determined entirely by the universal balance constant $\Lambda = \ln\varphi/\ln(2\pi)$ of the discrete-continuous transition. The cross-domain recurrence therefore does not require a physical relationship between ecology and nuclear structure; it requires only that both systems implement the same abstract information-profit condition, a condition that appears to be generic whenever a self-referential process must out-generate its dissipation costs in order to persist. Whether this constitutes a universal selection law or a family of structural analogies with a common fixed-point value is the empirical question the present paper is designed to test; the companion SRRG paper [4] provides the proposed theoretical derivation.

8.2 The Coxeter-conductor conjecture

Conjecture 8.1 (Coxeter-Conductor **[Conj]**)**.** *Under the axioms of locality, symmetry, and minimum description length, the minimal cyclotomic field containing all derived algebraic constants of a physical theory equals* $\mathbb{Q}(\zeta_N)$ *where* $N = \text{lcm}$ *of the Coxeter numbers of all symmetry groups selected by the theory. For the Standard Model,* $N = 120$.

The arithmetic backbone of this conjecture is fully Lean-certified (`BraidAtlas.CoxeterConductor`, `BraidAtlas.CoxeterConductorTowerLaw`, zero `sorry`, [3]). The field-theoretic embedding layer is also certified: for each algebra with $h \mid 60$, a $\mathbb{Q}$-algebra embedding CyclotomicField$(2h, \mathbb{Q}) \hookrightarrow$ CyclotomicField$(120, \mathbb{Q})$ is proved in `CyclotomicContainment` (`cyclotomic_field_embedding`, zero `sorry`, [3]). The E7 Tower Law exclusion is the Lean-certified structural non-containment (P24 Corollary [5]). The CKM and PMNS mixing angles do *not* lie in $\mathbb{Q}(\zeta_{120})$ (nor in any $\mathbb{Q}(\zeta_N)$ with $N \leq 360$): this is confirmed by PSLQ at 60 decimal digits across all eight mixing parameters [5]. The field-theoretic interpretation (why MDL selects exactly these Coxeter numbers) is a subject for future theoretical work.

8.3 Relation to the UGP

The two-stage sieve is not a prediction of the Universal Generative Principle itself. The UGP [6, 7] selects the Standard Model seed through the RSUC (Residual Seed Uniqueness at $n = 10$) and the GTE cascade, which constitutes one instance of the admissibility filter. The observation that this same two-filter structure appears in nuclear physics, biochemistry, and integrable field theory originated in the analysis of the UGP results in P01 [2] and was formalised in P24 [5]. The present paper synthesises the growing cross-domain evidence and proposes the GSP as a hypothesis about the universality of the mechanism.

8.4 Limitations and open problems

Several important questions remain open.

The IPT value. The value IPT ≈ 1.1309 is confirmed in ecology and economics (exact), and in nuclear physics at the 1.6% level. For the genetic code, the IPT connection is indirect (via the prebiotic model, which is indirect). Whether IPT is exactly 1.1309 or represents a band [1.05, 1.15] remains to be determined, and the deeper question of why this value is universal requires a more fundamental derivation.

A criticality interpretation. The IPT threshold may be related to a phase transition or critical point in the underlying dynamics. In statistical physics, the critical point of the 2D Ising model is the precise value at which the correlation length diverges; an analogous interpretation for the Gen/Drain renormalisation fixed point is a natural direction for future work. Recent independent work establishes that thermodynamic efficiency—the ratio of predictability gained to energy cost—is maximized precisely at the critical regime of the 2D Ising model [18], providing theoretical grounding for the claim that G/D is an efficiency-like selection pressure with a preferred fixed point above the Zipf baseline. Connections to other criticality thresholds in the literature (neural avalanche criticality, linguistic phase transitions) deserve investigation.

The algebraic conjecture for 3D Ising. The 3D Ising critical exponents are conjectured to be algebraic numbers. If they lie in $\mathbb{Q}(\zeta_N)$ for some N, this would extend the $\mathbb{Q}(\zeta_{120})$ evidence to statistical mechanics. Testing this conjecture requires bootstrap precision of at least 30 significant figures; current best results stand at 9 significant figures [19], and the rate of improvement is approximately one digit per three years. We note the conjecture is consistent with all currently available numerical data.

9 Conclusion

We have presented two-stage sieve evidence across five empirical domains — ecology and economics (original IPT derivation and confirmation, [C]/[B]), nuclear magic numbers ($\kappa/\kappa_{\min} \approx$ IPT, [Br]), prebiotic amino-acid chemistry (Stage 1 and Stage 2 selection, [Br]), organizational selection (three-regime failure cascade 6.81% → 1.11% → 0.22%, ΔBIC $= -92.19$, inception-cohort HR $= 5.01\times$, variance phase transition BC $= 0.907$; seven tests [B$^-$]), and microbial metabolic selection (*E. coli* iJO1366 FBA, $r = 0.971$, $p < 0.001$ [B]) — one model-internal symbolic-communication result (natural language alphabet, $n = 26$ unique fixed point, [B]), and one theorem-level algebraic substrate family ($\mathbb{Q}(\zeta_{120})$ with E7 falsifier, E8/ADE/WZW, Lean-certified [T]/[C]) — together with a speculative theoretical bridge to cognitive systems (Appendix D, [I]).

The quantitative recurrence of IPT ≈ 1.1309 and of $\mathbb{Q}(\zeta_{120})$ across these domains — with a clean algebraic falsifier in the E7 case — constitutes non-trivial evidence for the General Selection Principle.

The framework is falsifiable. Additional falsifiers would be: any ADE Toda theory with $h \nmid 120$ whose masses *do* lie in $\mathbb{Q}(\zeta_{120})$; any physical system at a confirmed IPT value substantially different from 1.1309; or bootstrap results for the 3D Ising model that are inconsistent with algebraicity. None of these falsifications has been observed.

Author Contributions

N.S. conceived the framework, performed all derivations, implemented the computational analyses, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

Code and data are available at `https://github.com/novaspivack/ugp-physics`. All numerical results supporting the claims in this paper are archived in the `papers/26_general_selection/data/` directory of that repository. The flux-balance analysis results underlying the microbial metabolic selection test (§6) are archived in `data/microbial_ipt_results.json`; the generating script (`scripts/microbial_ipt_test.py`) uses COBRApy 0.31 with the publicly available iJO1366 genome-scale metabolic model [20]. The Lean formalizations are at `https://github.com/novaspivack/ugp-lean`.

Computational Tools

Python 3 with NumPy and mpmath was used for PSLQ computations and the prebiotic fitness model. The machine-checked formalization (`ugp-lean`) is written in Lean 4 against Mathlib, with zero `sorry` and zero custom axioms; see companion formalization paper [21].

Full reproduction instructions are provided in `REPRODUCE.md` co-located with this paper's source.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

A Assumption Ledger

The following table lists the key assumptions and premises of this paper, together with their status in the UGP claim taxonomy.

Label	Statement	Status	Depends on
A-IPT	IPT $= 1 + \ln\varphi/(2\ln(2\pi)) \approx 1.1309$ is the unique fixed point of the Gen/Drain renormalisation group	[T] (conditional)	A1–A3 of P15 [1]; structural premises A2, A3 are physically motivated but not themselves derived from first principles
A-GTE	GTE pion-exchange parameters give spin-orbit coupling $\kappa = 0.050$ and $\kappa_{\min}(N{=}50) = 0.0435$	[Br]	Nuclear Physics (P03) [9]
A-SIPF	SIPF (salt-induced peptide formation) rate constants from Rode (1999) [12] are quantitatively applicable to prebiotic ocean conditions	[C]	Published experimental data; Rode 1999; Miller & Bada 1988
A-Tau	The template enhancement factor $\tau = 10$ for standard amino acids reflects aminoacyl-tRNA synthetase specificity in the context of the prebiotic fitness model	[Br] (bootstrapping)	Genetic code must exist to confer the advantage; result shows consistency, not causal priority
A-GSP	The General Selection Principle holds universally: every discrete combinatorial system in nature is shaped by the conjunction of an admissibility filter and an IPT viability filter	[Conj]	Supported by evidence from §2–6; Appendix D extends speculatively to cognitive systems; not yet formally proved

Label	Statement	Status	Depends on
A-Org	Corporate organizations with $G/D \geq$ IPT occupy a safety basin; the failure cascade 6.81% $\to$ 1.11% $\to$ 0.22% (Destruction/Watch/Safe) with ΔBIC $= -92.19$, inception-cohort HR $= 5.01\times$, and variance phase transition BC $= 0.907$ (Levene $p = 0.046$) confirms IPT as a genuine second structural boundary exhibiting both first-moment and second-moment phase-transition signatures	[B$^-$]	Three-threshold survival analysis: $N = 1{,}354$ complete-analysis company-year observations drawn from the EDGAR XBRL universe [15] (raw panel sizes: $N = 17{,}324$ for AUC/ROC, $N = 1{,}158$ inception cohort, $N = 12{,}741$ variance-phase transition); 5.15$\times$ Watch$\to$Safe odds ratio (Fisher $p \ll 0.001$); inception-cohort prospective test ($N = 1{,}158$, 63 events): HR $= 5.01\times$, Cox $p = 0.0003$, log-rank $p \approx 0$; AUC $= 0.781$ overall (Test C); activist targeting AUC $= 0.698$ (Test B); variance phase transition ($N = 12{,}741$): BC $= 0.907$, Levene $p = 0.046$, dip $p \approx 0$ (Test V); see §5
A-Consc	Consciousness requires a self-referential dynamical system maintaining $G/D >$ IPT	[I]	Theoretical bridge from IPT framework + NEMS programme [22]; empirically consistent with neural edge-of-chaos criticality and consciousness [23–26]; specific IPT value awaits direct empirical test

B Nuclear Magic Number Summary

The following table gives the spin-orbit coupling ratio $\kappa_{\text{emp}}/\kappa_{\min}(N)$ for each magic number, computed by `nuclear_magic_binding.py`. S_{2n} values are from the SEMF with Strutinsky shell correction.

N	Z	$S_{2n}(N)$ (MeV)	$S_{2n}(N+2)$ (MeV)	Drop (MeV)	$\kappa_{\min}$	$\kappa/\kappa_{\min}$	$\lvert\Delta/\mathrm{IPT}\rvert$
2	1	0.00	−1.70	1.70	0.020	2.500	121%
8	4	−2.00	−9.31	7.31	0.028	1.786	58%
20	9	−2.82	−6.06	3.24	0.032	1.563	38%
28	13	−2.27	−4.96	2.69	0.038	1.316	16%
50	22	−2.00	−4.32	2.31	0.0435	**1.149**	**1.6%**
82	37	−0.31	−2.28	1.97	0.048	1.042	8%
126	57	0.67	−1.09	1.76	0.049	1.020	10%

Table 8: Spin-orbit coupling ratios at magic numbers. IPT = 1.1309. The bold row ($N = 50$) is the unique magic number with ratio within 2% of IPT.

Note on SEMF validity: The SEMF underperforms for light nuclei ($N < 10$); the negative S_{2n} values for $N = 8$ are SEMF artefacts and do not affect the coupling-ratio argument, which is based on the Nilsson shell model parameters [9].

C Prebiotic Fitness Model Data

The following table gives fitness scores for all 30 amino acids in the model (20 standard + 10 non-standard), generated by `prebiotic_fitness_model.py`. $F_i = \mathrm{SIPF}_i \times C_{\mathrm{rel},i} \times \tau_i$; IPT viability threshold $= F_{\mathrm{min,std}}/\mathrm{IPT} = 1.326$.

Name	**Abbr**	**SIPF**	C_{rel}	τ	F_i	**S1**	**S2**	**Verdict**
Glycine	Gly	1.00	5.0	10	50.00	✓	✓	selected
Alanine	Ala	0.95	3.5	10	33.25	✓	✓	selected
Serine	Ser	0.85	2.0	10	17.00	✓	✓	selected
Glutamate	Glu	0.85	2.0	10	17.00	✓	✓	selected
Aspartate	Asp	0.88	2.5	10	22.00	✓	✓	selected
Threonine	Thr	0.78	1.5	10	11.70	✓	✓	selected
Valine	Val	0.75	1.5	10	11.25	✓	✓	selected
Lysine	Lys	0.75	1.0	10	7.50	✓	✓	selected
Proline	Pro	0.80	1.0	10	8.00	✓	✓	selected
Arginine	Arg	0.70	0.8	10	5.60	✓	✓	selected
Glutamine	Gln	0.72	0.7	10	5.04	✓	✓	selected
Asparagine	Asn	0.70	0.7	10	4.90	✓	✓	selected
Isoleucine	Ile	0.68	0.8	10	5.44	✓	✓	selected
Leucine	Leu	0.72	1.2	10	8.64	✓	✓	selected
Phenylalanine	Phe	0.65	0.6	10	3.90	✓	✓	selected
Histidine	His	0.62	0.6	10	3.72	✓	✓	selected
Tyrosine	Tyr	0.60	0.5	10	3.00	✓	✓	selected
Methionine	Met	0.60	0.4	10	2.40	✓	✓	selected

Name	**Abbr**	**SIPF**	C_{rel}	τ	F_i	**S1**	**S2**	**Verdict**
Cysteine	Cys	0.55	0.5	10	2.75	✓	✓	selected
Tryptophan	Trp	0.50	0.3	10	1.50	✓	✓	selected
D-Alanine	D-Ala	0.90	2.0	1	1.80	×	✓	fail-S1
β-Alanine	b-Ala	0.92	2.5	1	2.30	×	✓	fail-S1
Norvaline	Nva	0.71	0.6	1	0.43	×	×	fail-S1
AIB	AIB	0.45	0.3	1	0.14	×	×	fail-S1
Ornithine	Orn	0.68	0.5	1	0.34	×	×	fail-S1
Homoserine	Hse	0.60	0.4	1	0.24	×	×	fail-S1
Sarcosine	Sar	0.55	0.8	1	0.44	×	×	fail-S1
GABA	GABA	0.30	0.2	1	0.06	✓	×	fail-S2
DAB	DAB	0.25	0.15	1	0.04	✓	×	fail-S2
Pipecolic acid	Pip	0.20	0.1	1	0.02	✓	×	fail-S2

S1 = Stage 1 (structural admissibility); S2 = Stage 2 (IPT viability). Stage-1 exclusion reasons: D-Ala: wrong stereochemistry; β-Ala: wrong backbone geometry; Norvaline, AIB: structural isomers; Ornithine, Homoserine: metabolic intermediates; Sarcosine: N-methylation breaks peptide planarity. Full data: `prebiotic_fitness_table.csv`.

D Speculative Extension: Generative Cognitive Systems

[I] — This appendix presents a theoretical bridge; no direct empirical test of the IPT value in cognitive systems has been performed.

A *generative cognitive system*—understood minimally as a finite, self-referential dynamical system capable of sustaining an internal model of its own state—faces a two-stage sieve analogous to the GSP. Stage 1 (admissibility) requires the system to be genuinely self-referential: it must contain within itself a representation of its own generative rules (cf. the fixed-point condition $S = L(S)$ of [27]). Open-loop control systems, thermostats, and simple automata are excluded. Stage 2 (viability) requires that the generative-to-dissipative rate ratio satisfies $G/D >$ IPT; systems below this threshold undergo informational collapse in which internal models degrade faster than they are replenished [1].

The MFRR/NEMS research programme formalises this structure. The Theorem of Foundational Finality (`NemS.foundational_finality`, zero `sorry`, [22]) establishes that any finite, self-knowing system must obey a fixed-point consistency condition; the Reflexive Closure Theorem (`ReflexiveClosure`, [22]) provides the Lean-certified backbone for the claim that reflexive systems form a closed class. The IPT threshold thus marks the boundary between self-referential systems that can sustain coherent internal representations (Regime 3, $G/D >$ IPT) and those that cannot (Regimes 1–2).

Note that relay and transduction channels (systems that transmit rather than generate information) operate at $G/D \approx 1.0$; the IPT marks the threshold specifically for *generative* awareness loci, not for neural transduction in general [26, 28].

Several independent empirical lines are consistent with a criticality threshold for consciousness, though they do not directly test the IPT value ≈ 1.1309: loss of consciousness under general anaesthesia occurs at sharp physiological thresholds consistent with a viability transition [29]; the brain operates near the edge-of-chaos critical point during conscious states and departs from this point under propofol/xenon anaesthesia, seizures, and vegetative states [23, 24]; and a systematic review of 71 publications confirms that departure from neural criticality correlates with unconscious states [25]. A rigorous test of the IPT-consciousness connection would require measuring G/D ratios in conscious and non-conscious systems — an experiment currently beyond available precision. We record this as a theoretical prediction for future investigation.

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